10. The crystallisation of sodium ethanoate from a super-saturated solution is used to release heat in reusable hand warmers.

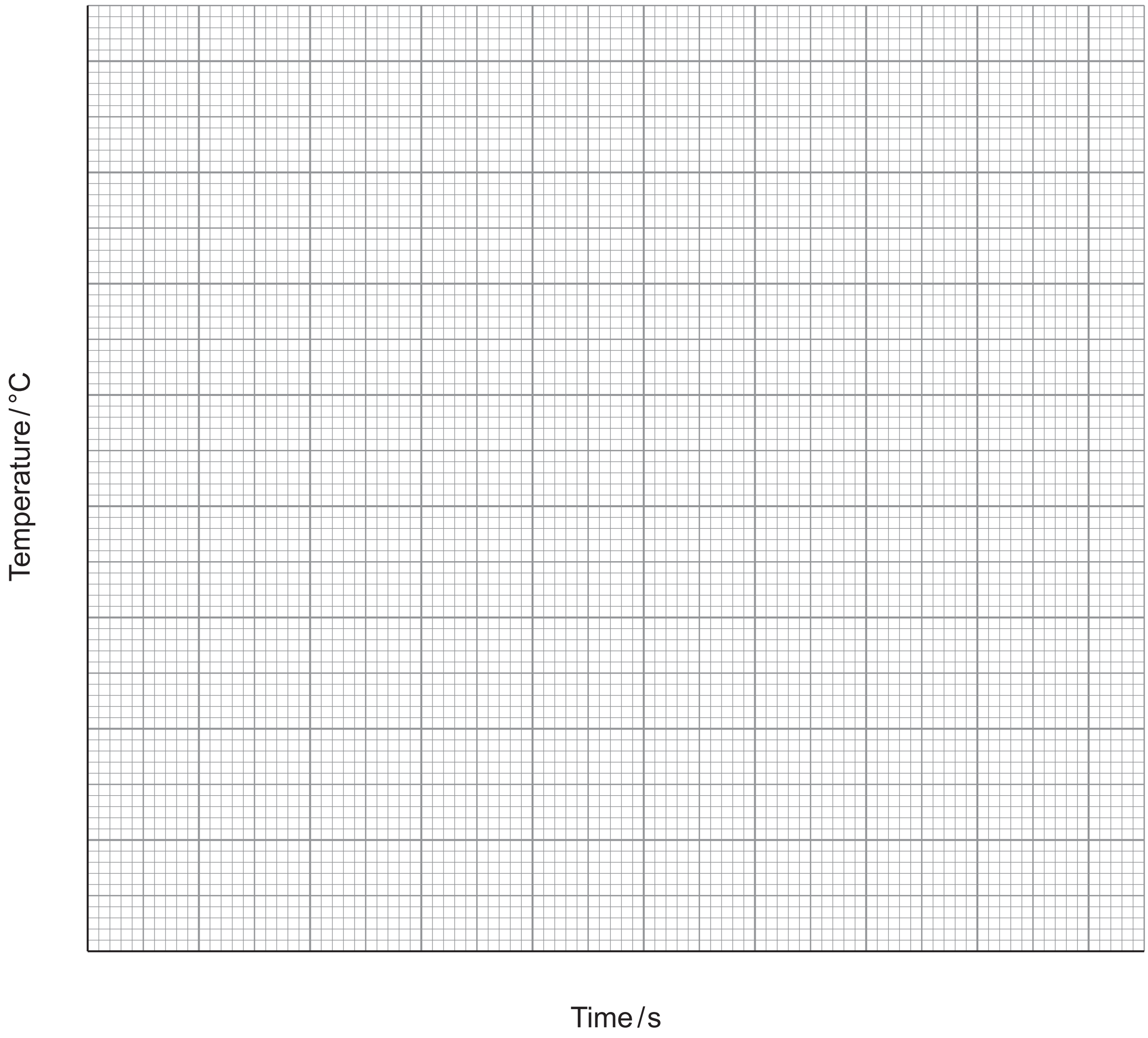
(a) A super-saturated solution of sodium ethanoate was made by dissolving 320 g of hydrated sodium ethanoate ($CH_3COONa.3H_2O$) in 60 cm^3 of hot water. It was then allowed to cool to room temperature, which was measured as 17 °C.

A thermometer was added to the solution, which caused the sodium ethanoate to start crystallising. The temperature of the process was recorded every 30 seconds for 3 minutes. The results are shown below:

Time / s	Temperature / °C
0	17
30	27
60	35
90	41
120	40
150	39
180	38

Examiner only

(i) Plot the results on the graph paper below. [2]



(ii) Use your graph to calculate the maximum temperature change for this crystallisation. [2]

maximum temperature change = ... °C

Examiner only

(iii) Use the **total mass** of the sodium ethanoate solution and the temperature change from the graph to calculate the enthalpy change of crystallisation per mole of sodium ethanoate. Assume the density of water is 1.00 $g\,cm^{-3}$ and the specific heat capacity of sodium ethanoate solution is 4.18 $J\,K^{-1}\,g^{-1}$.

$M_r(CH_3COONa.3H_2O) = 136$ [4]

enthalpy change = ... $kJ\,mol^{-1}$

(iv) Suggest a reason why the experimental enthalpy change is often lower than the theoretical enthalpy change. [1]

....

....

(b) Sodium ethanoate can be made in a neutralisation reaction. Complete the following equation: [2]

.... + ... $\longrightarrow$ CH_3COONa + CO_2 + H_2O

(c) The carboxylic acid used to produce sodium ethanoate can be produced using an oxidation reaction.

(i) Name the reagents and give the expected observations. [2]

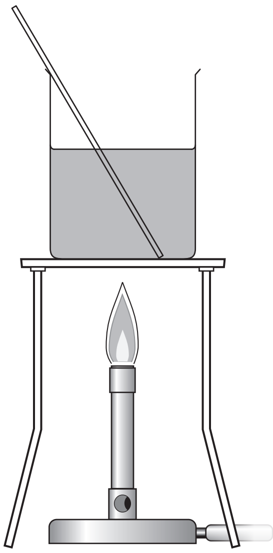
....

....

....

Examiner only

(ii) A student proposed that the apparatus below should be used to perform this oxidation reduction experiment.



The teacher said that this would not work and would be unsafe. Draw a labelled diagram of the apparatus that should be used in this experiment. [3]

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END OF PAPER